

Chicago Crime Patterns (2001- Present)

Group 5 (STAT 405)

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I: Introduction to Data

Crime is a persistent part of society, affecting people in many ways, whether directly as victims or indirectly through the communities they live in. This motivated our focus on Chicago crime, since Chicago is often recognized as one of the most violent cities in the United States, and we wanted to examine important trends and explore data that could contribute to public safety.

In this project, we use a reliable open source, the Chicago Data Portal, which contains incident-level records for crimes reported to the Chicago Police Department from 2001 - present. Each row represents one reported incident and includes information such as the date/time, location, crime classification (Primary Type, Description, IUCR, FBI Code), and binary fields. Location is measured by latitude and longitude, and geographic fields contain beat, district, ward, and community area. Binary fields include arrest and domestic outcomes.

Our primary question is:

How have reported crimes in Chicago changed since 2001 in terms of overall volume, time-of-day patterns, geographic concentration across community areas and police districts, and the relationship between crime type and arrest rates?

To further investigate these patterns, we also use a secondary data set containing police district and beat-level information, where beats represent smaller subdivisions within each district. By adding the secondary data set to the primary data set, there is a more detailed spatial aspect to the analysis as it allows pattern comparison across policing boundaries. Therefore, it is possible to identify which beats and districts experience heavier crime concentration.

The most important variables in this project are: Date, Time, Primary Type, Location, Arrest, and Domestic. The incident Date and Time are used to extract the Hour, Month, and Year. The Primary Type field is used for yearly trend plots and heat maps. The Location fields are used for spatial concentration and neighborhood and district-level analysis. The Arrest and Domestic indicators are used to analyze arrest outcomes and differences between domestic and non-domestic incidents.

To ensure that the data is ready for analysis, several transformations were applied. For summaries, data is restricted to specific year ranges depending on the plot. Year and count-related fields were converted into numeric form where needed for plotting and modeling. Furthermore, incident-level records were aggregated into counts across different units of time or geography, in order to illustrate the patterns more clearly.

II: Long-term Overview of Crime Trends

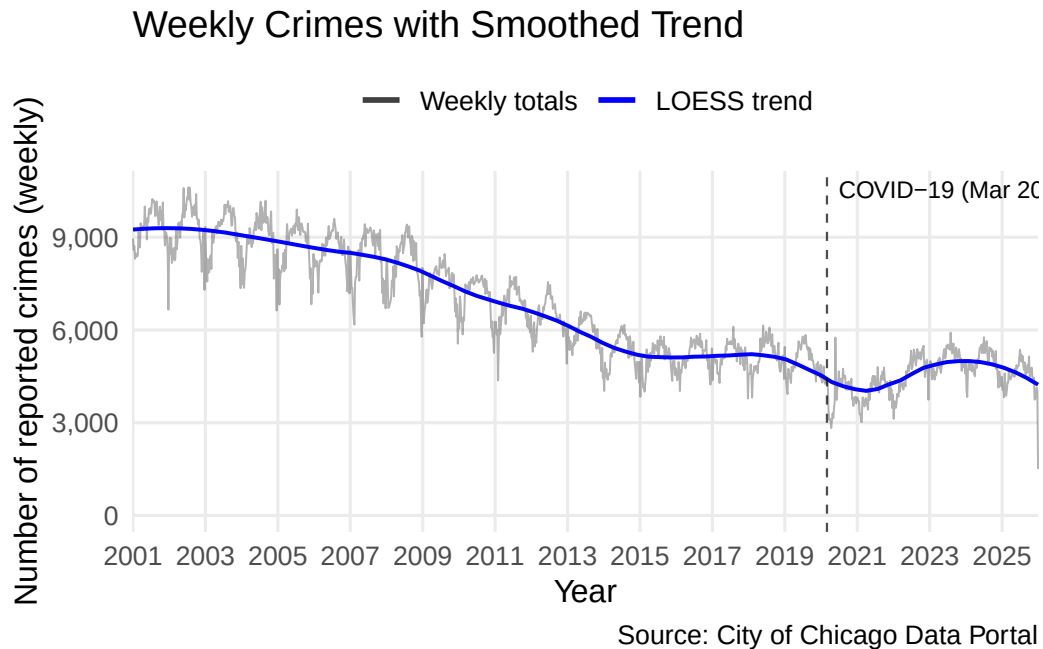


Figure 1: Time Series

The weekly totals plot with a LOESS smoother shows a strong long-run decline in reported crime from 2001 through roughly the mid-2010s, followed by a flatter period and then a noticeable dip around 2020–2021 near the COVID-19 marker. After COVID-19, there is a slight rebound followed by a slight decline in 2025. Looking at the weekly totals, there is a clear cyclical pattern throughout the years, illustrating that middle months have higher crimes and early and late months have less crimes. Taken together, these plots suggest that Chicago crime levels are shaped by both slow structural trends over decades and predictable yearly seasonal cycles, with major societal disruptions (such as COVID-19) capable of producing temporary deviations from the typical pattern. This motivates later analysis by crime type and seasonal effects rather than relying on overall totals alone.

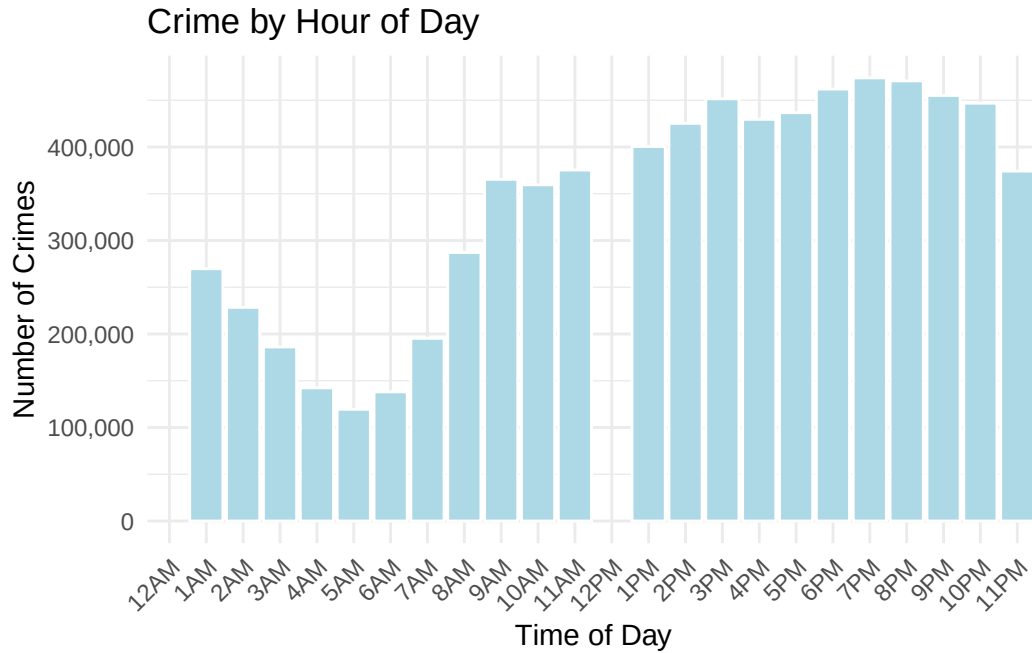


Figure 2: Histogram of Crime By Hour

To understand when reported crimes tend to occur during a typical day, we first examine the time-of-day distribution of incidents. Using the incident time stamps, we extract the hour (0–23) for each report and visualize the frequency of crimes across the 24-hour cycle. This helps identify peak hours and quieter periods. This histogram shows the average daily cycle of reported crimes, with the lowest activity roughly around 4am-6am, and highest activity at 3pm and 6-9pm. Crime activity gradually rises throughout the morning hours, and gradually declines after about 6pm. For this plot, 12am and 12pm data were excluded since it seemed to be unusually high perhaps being defaults when a specific time was not available.

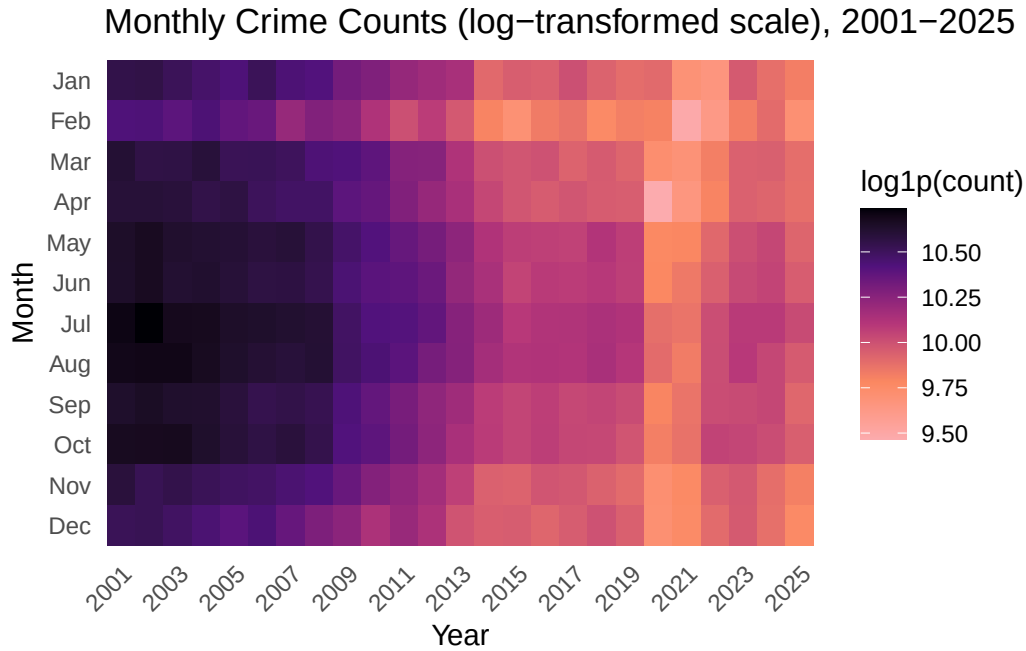


Figure 3: Heat Map

This is a heatmap showing the distribution of crime throughout the 12 months of a year for years 2001 to 2025. Lighter pinks represent months with lower levels of recorded crime and darker magentas and purples show higher rates of recorded crime. On first glance, it is evident that there is a gradual decline throughout the years, with a notable disruption in 2020-2021 due to COVID-19, which supports the findings of the time series plot. A log-transformed color scale is used to improve contrast between months and years, since raw counts would compress the variation into a narrow range of colors. The trend of crime oscillation throughout the year from figure 1 is clearer here as it can be seen that crime increases in the middle of the year, during the warmer months. It also appears that February and to some degree, March, tend to have the lowest crime rates out of the entire year. Chicago is known to have extreme winters that are life-threatening to anyone without shelter and constant access to heating. This is an environmental condition that would make it more difficult for certain types of crimes to occur (theft, robbery, assault, battery, burglary, to name a few).

Chicago Monthly Crime Count vs. Temperature (2001–2025)

Monthly crime counts and average monthly temperature

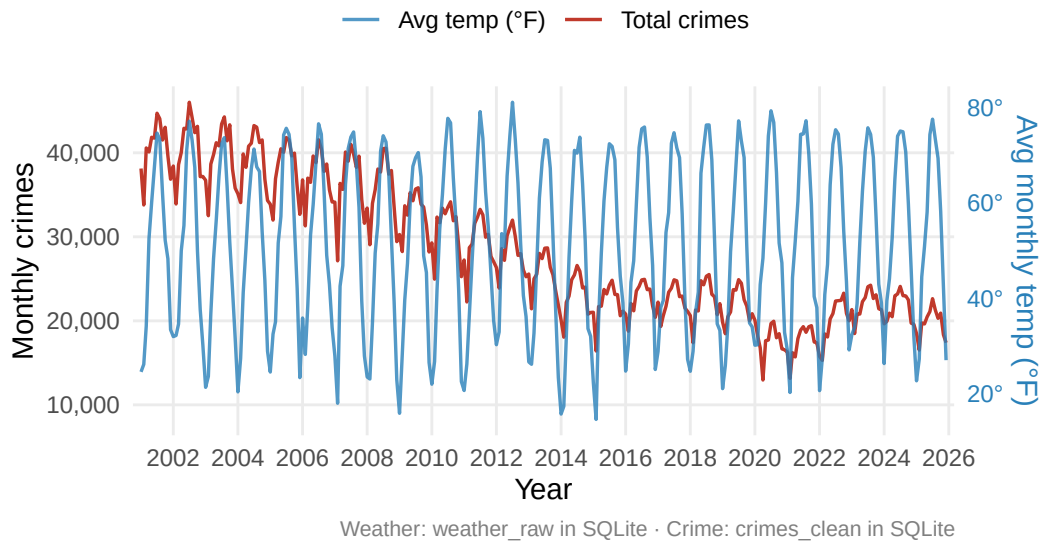


Figure 4: Monthly Crime Count vs Temperature Plot

The curve showing monthly crime totals over the course of 2001-2025, exhibits the same up, down, then up again tendency of the weekly crime totals curve in the plot with the LOESS smoother. Interestingly enough, this pattern aligns almost completely with the alike tendency (up, down, then up again) in the average monthly temperature curve, showing that characteristic yearly increases in crime coincide almost perfectly with yearly increases in temperature as it goes to the middling months of the year, and then down again as it goes to the end months. However, the overall trend of crime over the years (like the weekly crimes plot with the LOESS smoother) is a downward slope, whereas the weather looks as though it has averaged about the same temperature distribution over the observed span of 25 years. The characteristic movement of crime over the months in the year still follows the same pattern over the entirety of the period (not breaking this cycle), almost completely aligned with the average temperature curve, though this long-term downward trend is not seemingly influenced at all by the weather trends. There is also, in the years 2020 to 2022, that same dip in the overall trend of crime that was found in the weekly crimes plot, most likely from the COVID-19 pandemic, followed by the correction of the overall trend in crime in the years after. Overall, the trend of crimes yearly is that the change from the lowest point to the highest point in the characteristic yearly cycle (change from lowest to highest point in each wave) has gotten smaller over the 25-year period. Though with close and careful observation, this can be seen too in the weekly crimes plot. Consequently, it seems that average temperature, if it has any effect on this tendency, probably has a minuscule effect upon it.

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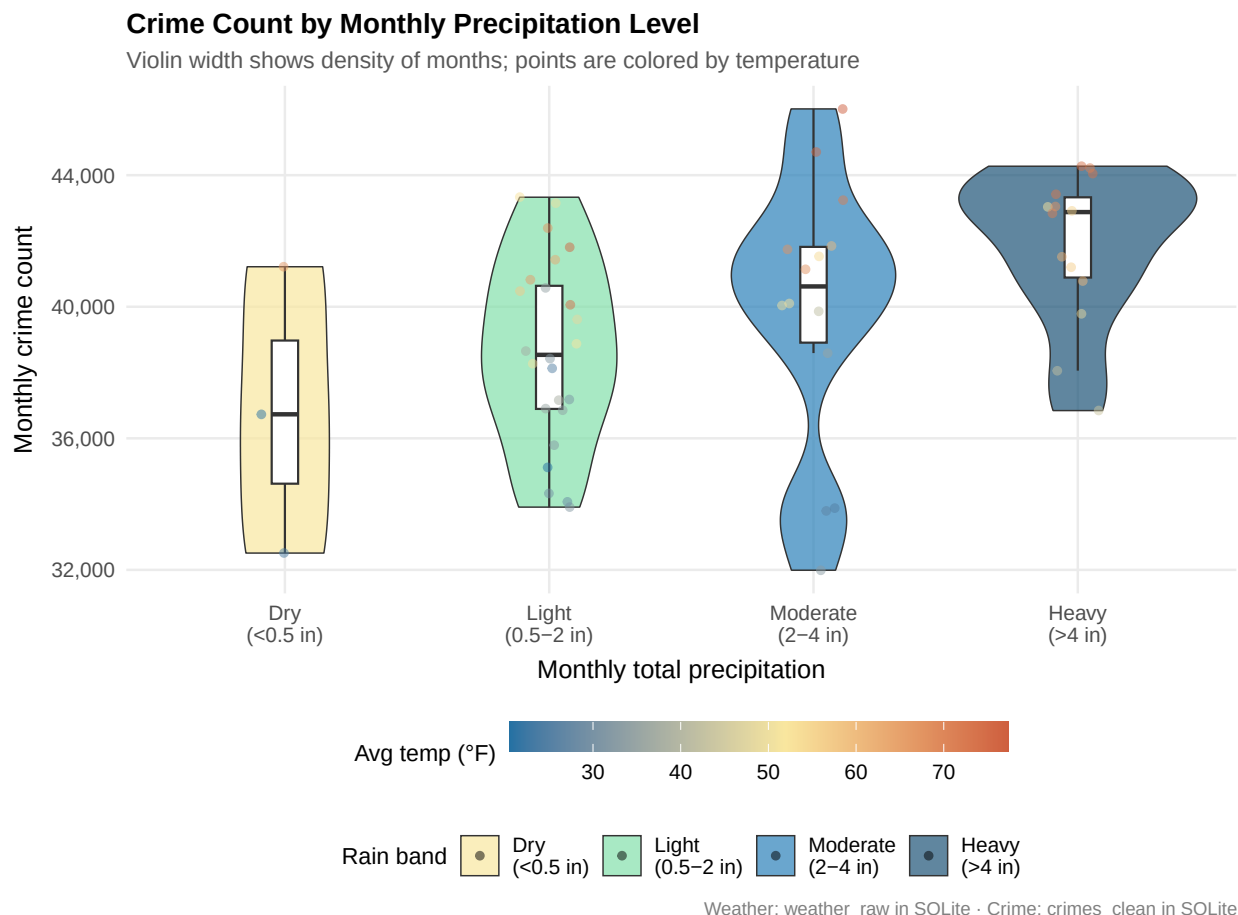


Figure 5: Weather Violin Plot

The graph shows monthly crime totals across four precipitation groups: “Dry” (less than 0.5 inches), “Light” (0.5–2 inches), “Moderate” (2–4 inches), and “Heavy” (more than 4 inches). Each point is colored by average monthly temperature, which helps show whether temperature may be influencing the pattern more than precipitation itself. The boxplots show medians and interquartile ranges, while the violin widths show where monthly crime totals are most concentrated within each rainfall group.

At first glance, the plot seems to suggest what one might expect: crime decreases as rainfall increases. This impression comes mainly from the medians, since the median monthly crime total for “Dry” months is higher than the medians for the wetter categories. However, that conclusion may be misleading for several clear reasons. First, there are relatively few “Dry” months, so that group may not be very reliable for comparison. Second, the medians for the “Light,” “Moderate,” and “Heavy” groups are fairly similar, with “Heavy” months even appearing slightly higher than “Light” months. Third, the interquartile ranges do not differ much across groups, suggesting that the central distributions are more alike than the medians alone imply. Taken together, these features weaken the idea that higher monthly rainfall clearly deters crime.

Another noticeable feature is the shape of the violins. In the “Light” and “Heavy” groups, the density appears to cluster near both the lower and upper ends of the monthly crime range, while the “Moderate” group is more concentrated toward the lower end. This likely reflects the broader long-term decline in crime over time rather than a simple rainfall effect. Earlier years contribute many higher monthly crime totals, while

later years contribute much lower ones, creating a split distribution within these precipitation categories. Because of this, the overall downward trend in Chicago crime may be affecting the rainfall comparison in a way that makes any deterrent effect of precipitation difficult to isolate.

III: Crime-Type Analysis

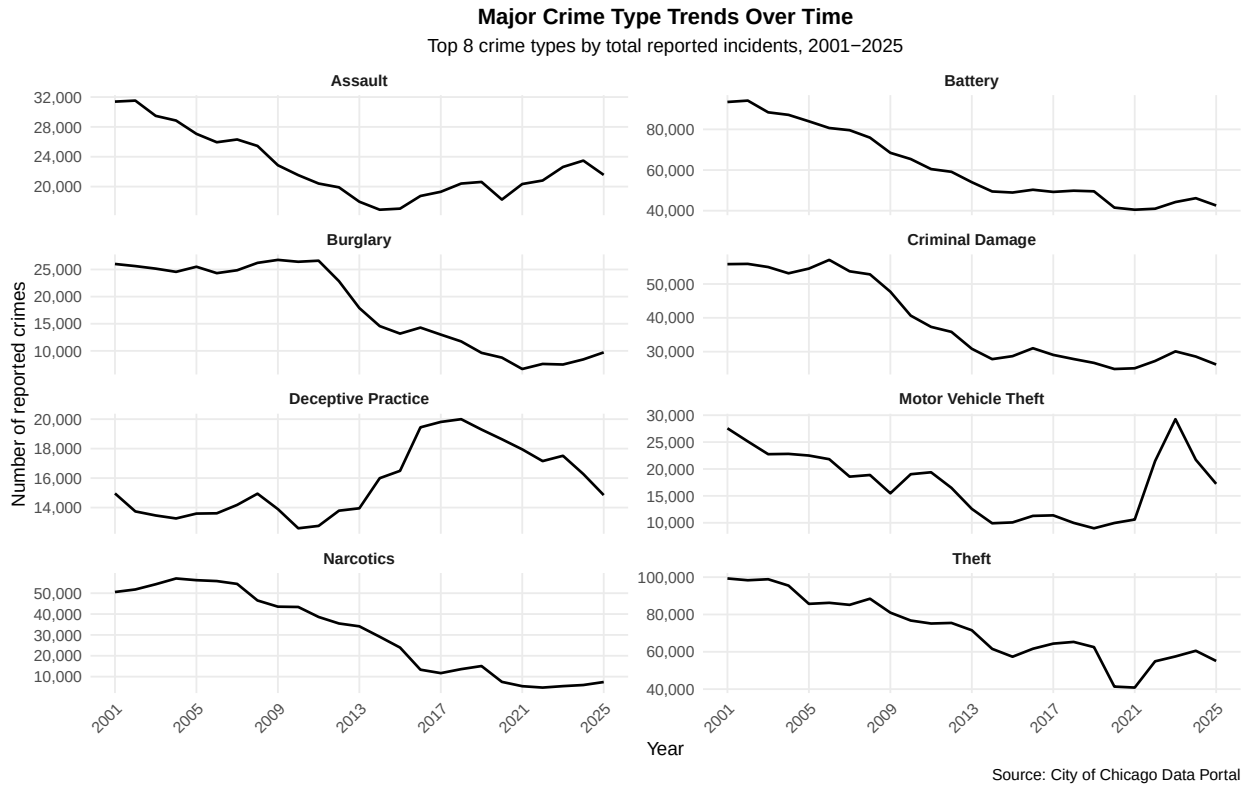


Figure 6: Top Crime Types

These are the top eight crime types excluding the “Other Offense” category with each graph measuring reported crime counts over time. Deceptive practice, narcotics, battery, and criminal damage are decreasing in recent years, while assault, burglary, motor vehicle theft, and theft have some higher counts in recent years. There are several spikes in the graphs of deceptive practice and motor vehicle theft. This motivates a deeper analysis of crime by hour of day.

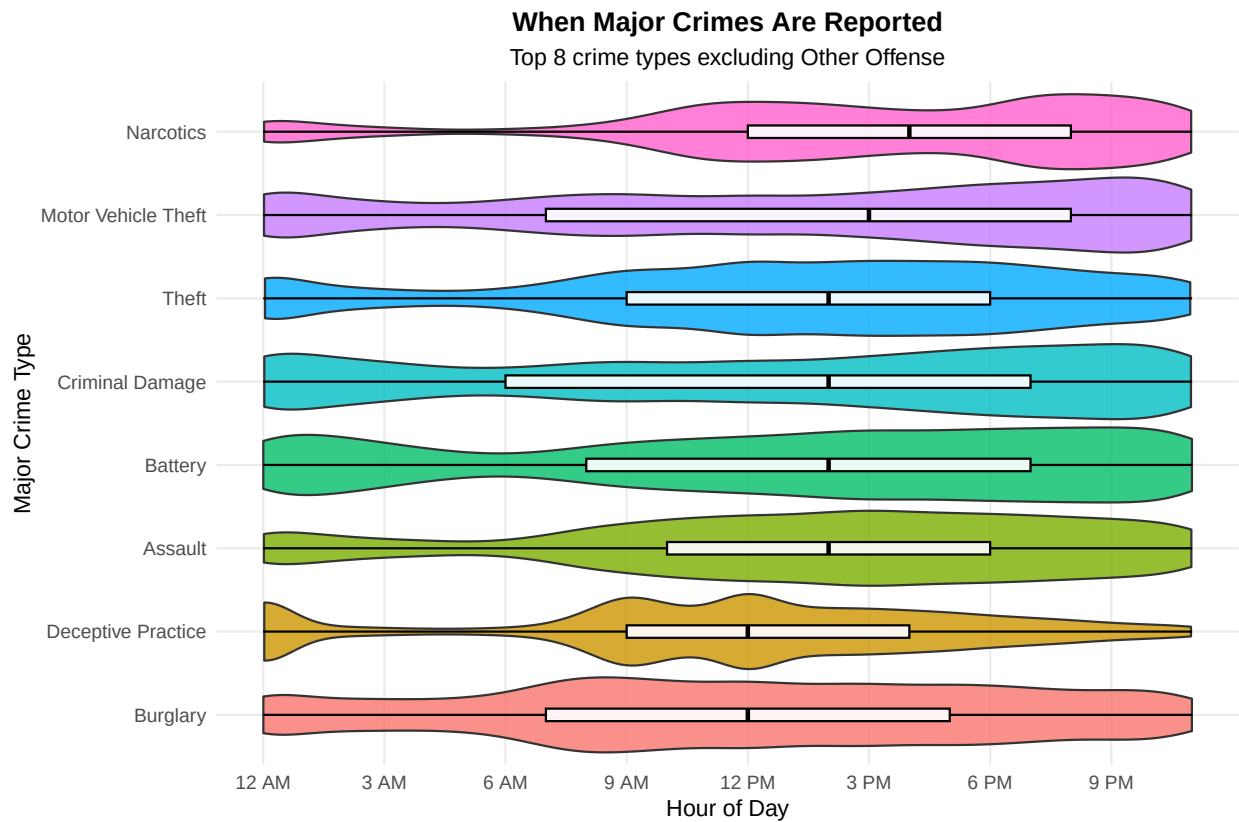


Figure 7: Violin Plot

In addition to how often a crime occurs, it is worth examining when it is most likely to be reported throughout the day. This plot shows the distribution of report times for the eight most common crime types, excluding the Other Offense category. The width of each plot represents reported crime count and the box inside each plot shows the median and interquartile range. The crimes are ordered by median report hour which highlights the time differences among crimes. Narcotics immediately stands out because its density is strongest later in the day and very weak in from 3am-6am. Deceptive practice is also unique as its density is very weak from around 2am-6am, while having a distinct midday-centered pattern with the tightest central box out of the crime illustrated. Other crimes such as motor vehicle theft, theft, criminal damage, battery, assault, and burglary, seem to have more consistent densities with most peaking in the evening hours.

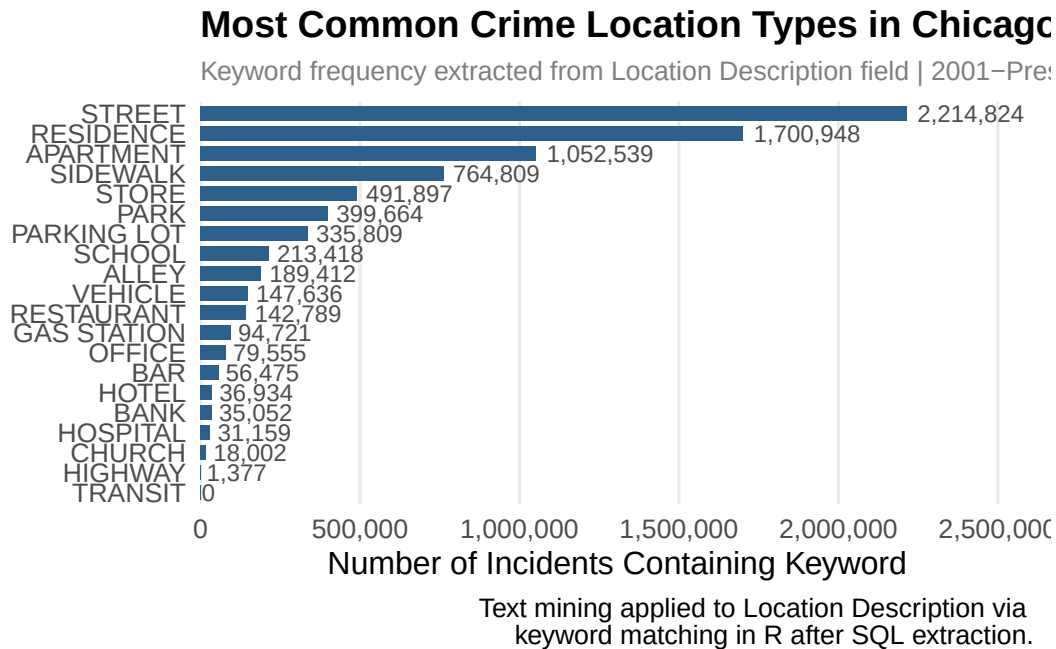


Figure 8: Text-Mining Bar Chart

The bar chart displays the frequency of key location-type keywords extracted from the Location Description field of the Chicago crimes dataset, spanning from 2001 to the present. A text mining approach was applied by scanning every entry for general place-type keywords such as “street,” “residence,” “apartment,” “sidewalk,” and others, allowing granular and varied descriptions to be rolled up into broader, more interpretable categories.

The results reveal that streets and residences are by far the most common settings for reported crimes in Chicago, suggesting that a significant portion of criminal activity occurs either in open public spaces or within or immediately around private homes. Sidewalks and parking lots also rank highly, reinforcing the pattern that outdoor and semi-public spaces represent elevated risk environments. More enclosed or specialized locations such as banks, hospitals, and churches appear far less frequently, which may reflect both lower foot traffic in those settings and stronger security deterrents. Schools, parks, and gas stations fall in the middle range, indicating moderate but consistent criminal activity in those areas. Taken together, these findings suggest that crime prevention efforts in Chicago may be most impactful when concentrated on street-level and residential environments, as these locations account for the overwhelming majority of reported incidents.

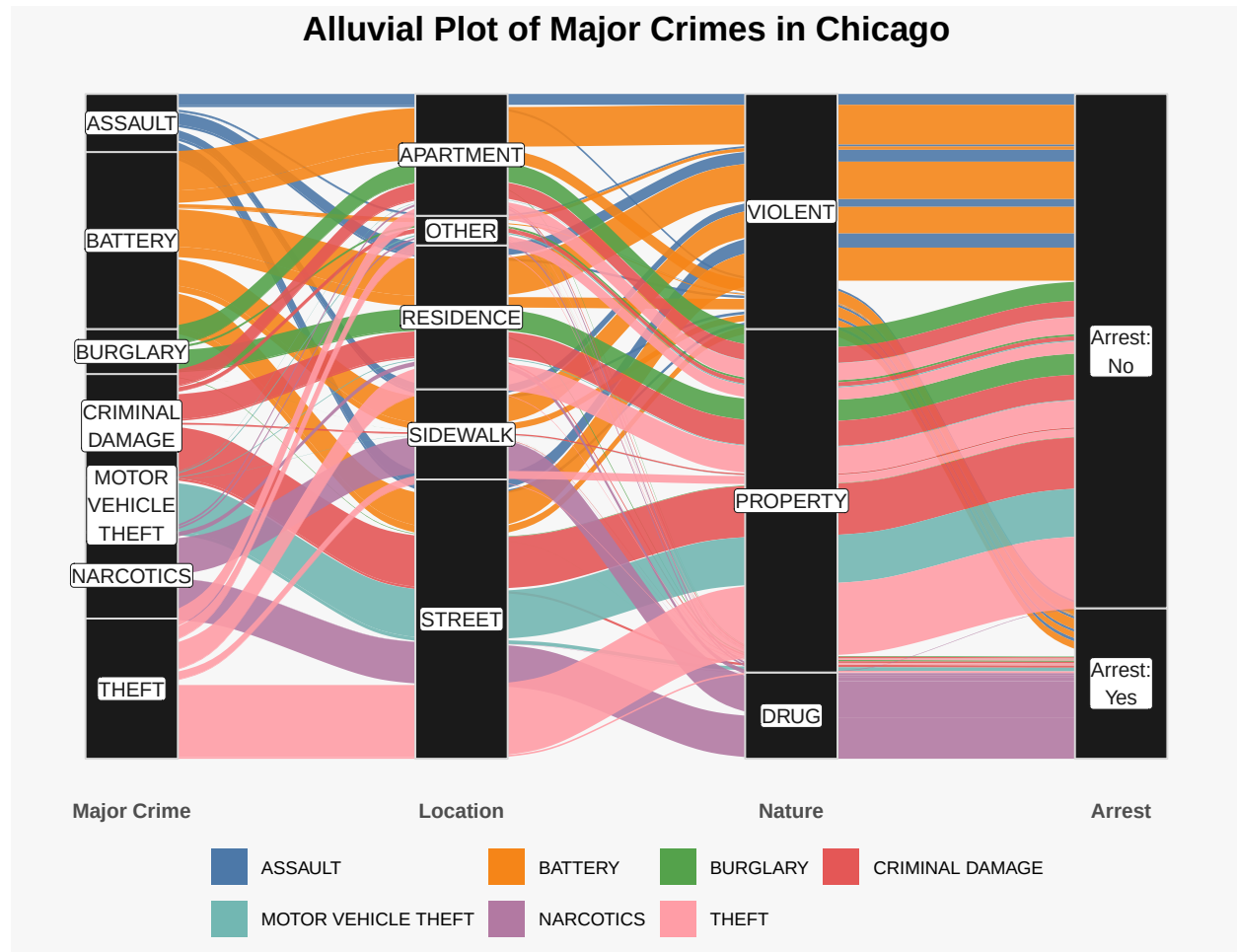


Figure 9: Alluvial Plot

People generally believe that committing a crime will lead you to being arrested. It turns out, at least in Chicago, that a significant portion of crimes never result in an arrest, whether that be due to lack of sufficient evidence, lack of resources, too much bureaucracy, or corruption. This alluvial plot shows the relationship between the most committed crimes in Chicago (which each have over a hundred thousand instances over the 2001-2025 time period), the nature/classification of those crimes, the most common locations with criminal activity, and whether or not those crimes resulted in an arrest being made by Chicago PD. Narcotics, and by extension drug crimes, almost always result in arrests. They mostly occur outside (on the street or the sidewalk) instead of within residences or apartments. There is not any reason to believe that narcotic/drug crimes occur outside more frequently than inside; generally, people hold the notion that most drug use occurs within one's own home. So, this trend could be indicative of the higher probability of being caught when performing a drug crime outdoors versus indoors. The vast majority of property crimes seem to not result in an arrest whilst more violent crimes do proportionally result in an arrest (although the majority of violent crimes also don't result in arrests). Of the violent crimes that do result in an arrest, they are mostly battery charges. This may be because battery is the actual physical act of harming another whilst assault is merely the threat to do so. Of the property crimes that do result in an arrest, the split is roughly equal between criminal damage (vandalism) and motor vehicle theft. Future work that follows this model could look at more locations to build a stronger plot that could help to visualize whether certain locations have any impact on the type of crime that occurs there and whether or not arrests happen there.

Table 1: Logistic Regression Fit Statistics for Arrest

Metric	Value
AIC	1526799.7
Null deviance	2091014.94
Residual deviance	1526755.7
Likelihood ratio chi-square	564259.24
Model df	21
Residual df	2516600
Model p-value	0e+00

Table 2: Logistic Regression Coefficients for Arrest

Term	Log-odds	Std. Error	z-statistic	p-value
Intercept	126.2122	1.3744	91.8334	0.0e+00
Hour Of Day	0.0065	0.0003	20.1447	3.0e-90
Year	-0.0634	0.0007	-93.2097	0.0e+00
Domestic Incident	0.0247	0.0132	1.8696	0.0615
Battery	0.1884	0.0092	20.4683	4.1e-93
Burglary	-1.1750	0.0157	-74.9156	0.0e+00
Criminal Damage	-1.3825	0.0123	-112.4657	0.0e+00
Deceptive Practice	-1.4864	0.0141	-105.2788	0.0e+00
Motor Vehicle Theft	-1.1933	0.0140	-85.0929	0.0e+00
Narcotics	6.5251	0.0337	193.6797	0.0e+00
Other Offense	0.6446	0.0102	63.4456	0.0e+00
Robbery	-0.7044	0.0139	-50.6954	0.0e+00
Theft	-0.6231	0.0087	-71.9353	0.0e+00
Domestic $\times$ Battery	0.1950	0.0151	12.8801	5.8e-38
Domestic $\times$ Burglary	0.9064	0.0648	13.9976	1.6e-44
Domestic $\times$ Criminal Damage	0.8940	0.0233	38.4055	1.1e-322
Domestic $\times$ Deceptive Practice	-0.7751	0.1549	-5.0044	5.6e-07
Domestic $\times$ Motor Vehicle Theft	0.6328	0.0735	8.6065	7.5e-18
Domestic $\times$ Narcotics	-2.4320	0.3054	-7.9626	1.7e-15
Domestic $\times$ Other Offense	-0.9854	0.0194	-50.7718	0.0e+00
Domestic $\times$ Robbery	0.1607	0.0746	2.1531	0.0313
Domestic $\times$ Theft	-0.7229	0.0319	-22.6679	9.3e-114

Statistical Modeling: Regression

Purpose of the Model

The purpose of this model is to understand which contextual and temporal factors significantly affect the probability that a crime results in an arrest.

A logistic regression model was used, where the response variable Arrest01 is coded as 0 or 1. This model estimates how each predictor is associated with the probability that an incident results in an arrest. To reduce the number of categorical levels and improve interpretability, the analysis restricts Primary_Type to the top 9 most frequent crime types in the data since 2015, while also including Domestic, Hour, and Year as predictors.

The fit-statistics table summarizes overall model performance. The coefficient table reports estimated effects for each predictor. In this logistic model, a positive coefficient indicates that the predictor is associated with higher odds of arrest, while a negative coefficient indicates lower odds of arrest, holding the other variables constant. For example, the coefficient on Domestic represents the difference in arrest odds between domestic and non-domestic incidents, while the coefficients on Hour and Year represent the change in arrest odds for a one-unit increase in hour or year. The interaction terms show how the effect of domesticity changes depending on crime type. In general, domestic incidents appear more likely to result in arrest, although this pattern is weaker or reversed for deceptive practice, theft, and especially narcotics. Further sociological research could help explain why domesticity in these categories is associated with lower odds of arrest.

These associations should not be interpreted as causal. Arrest outcomes can be influenced by unobserved factors such as the presence of a suspect, reporting behavior, and policing resources, and some predictors may also be correlated with one another. Additionally, because the model focuses on the most common crime types, the results describe arrest patterns for high-frequency categories rather than the full set of offenses. More advanced modeling, such as logistic regression with regularization or neighborhood controls, could further clarify which factors are most predictive of arrest and how those relationships vary across contexts.

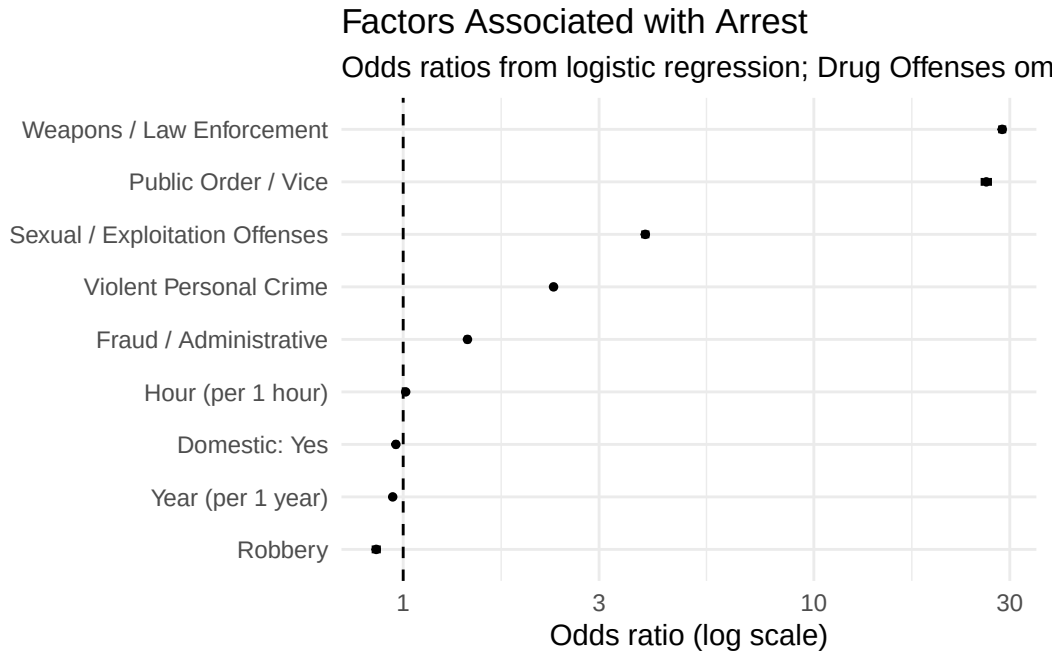


Figure 10: Coefficient Plot

To examine which factors are most strongly linked to whether an incident leads to an arrest, a logistic regression model is fitted where the outcome is Arrest (yes/no). Crime categories are grouped into eight main crime groups to focus on substantive patterns rather than just raw frequency rankings. This also helps interpretability and broader coverage of the data.

The predictors include Crime Group, Domestic (yes/no), and time variables, Hour and Year. The plot illustrates the model results as odd ratios with confidence intervals on a logarithmic scale, making it easier to compare factors linked to higher arrest odds versus lower arrest odds.

In this plot, the dashed vertical line at odds ratio = 1 represents no difference in arrest odds. Points to the right of 1 mean higher arrest odds, while points to the left of 1 indicate lower arrest odds, holding the other predictors constant. The reference category is Theft/Property, so all other coefficients are compared relative to Theft/Property, holding the other variables fixed. The primary correlation is that crime type is the key driver of arrest probability. Relative to Theft/Property crimes, offenses in the Weapons / Law Enforcement and Public Order / Vice categories have the highest odds of arrest, while Sexual /Exploitation Offenses and Violent Personal Crime are also associated with elevated arrest odds. In contrast, Robbery is associated with lower odds of arrest than the reference group. This is likely because robbery is often difficult to be cleared through arrest due to fast movement, limited identifying information, and fewer immediate leads. The effects of Hour, Domestic status, and year are comparatively small, suggesting that crime group is the strongest predictor of arrest in this model.

IV: Community-Level Analysis

Average Annual Crime Rate by Community Area (2020–2024)

Average yearly crimes per 1,000 residents

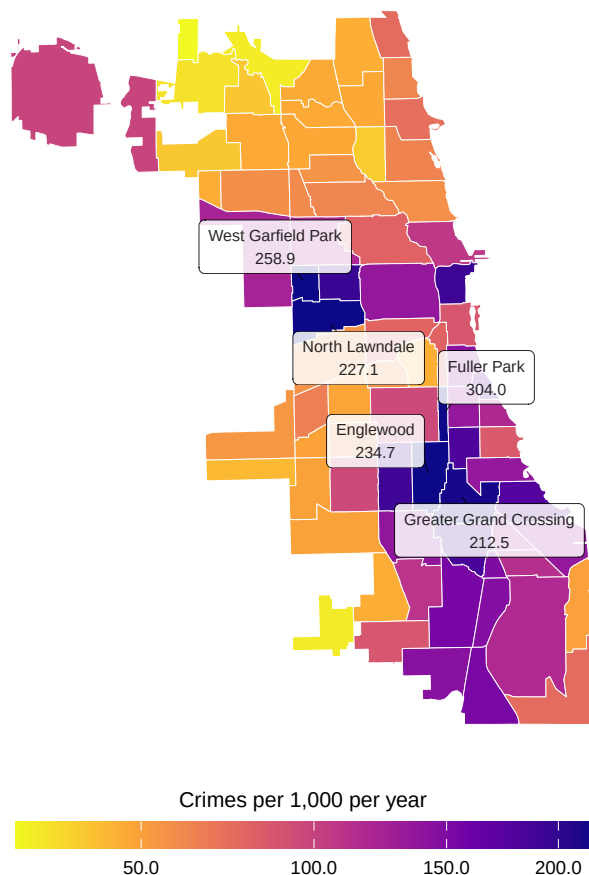


Figure 11: Choropleth

Raw crime counts can be misleading because community areas vary widely in population size. To make comparisons better, this figure maps an average annual crime rate for each community area from 2020–2024. This is computed by dividing the area’s average yearly number of reported incidents by its ACS 2023 residential population, expressed per 1000 residents. This per-capita approach highlights where crime is most frequent relative to the number of residents rather than where the most incidents happen in total. The map reveals a high geographic variation in crime rate across Chicago, even after normalizing by population. Lighter (orange/yellow) areas cluster in lower-rate ranges, while darker (purple) areas indicate higher rates. Several community areas stand out as hot spots, labeled on the map, suggesting that elevated per-capita crime rate is concentrated instead of uniformly distributed across the city during 2020–2024.

However, these patterns should not be interpreted as purely risk or danger. Several confounding factors can impact rates in certain areas including population density, transit hubs, and nightlife. Furthermore, difference in reporting behavior and policing practices across police districts can also affect recorded incident counts. Therefore, this map does not prove a causal explanation of higher crime rates in some areas. Methodologically, a five-year average is used which reduces noise from annual fluctuations. The color scale is capped at the 95th percentile, which prevents extreme values from hiding variation and compresses the areas with the highest rates. Overall, this map gives a clear starting point for deeper analysis, particularly breaking rates down by crime type and comparing time periods.

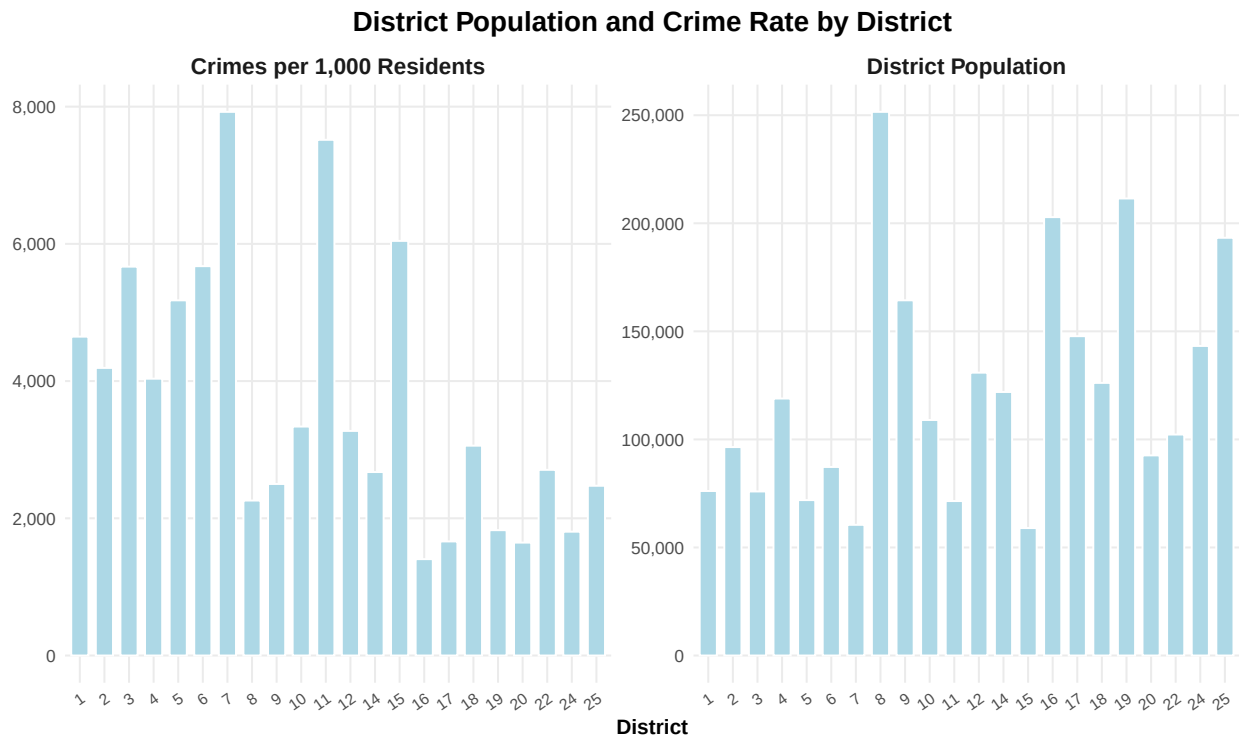


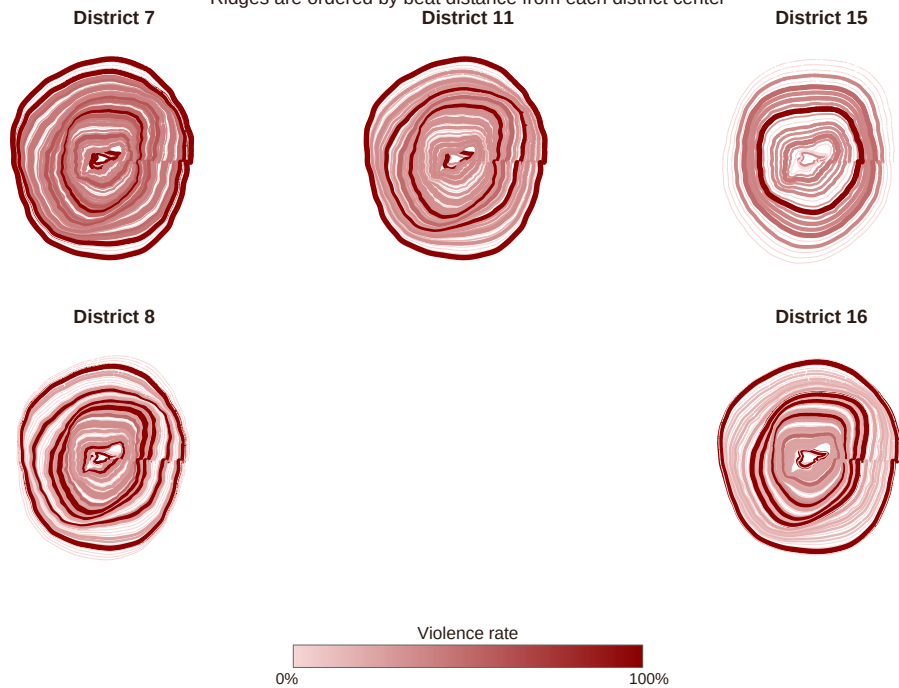
Figure 12: Police Districts Bar Charts

Based on these plots, it is evident that police districts have a wide range of crime intensity. The first plot shows the crime rate for each district and the second plot shows each district's population. From the first plot, Districts 7, 11, and 15 have the highest crime rates and have some of the lowest populations. On the other hand, Districts 8 and 16 have low crime rates and higher populations. This may be due to several factors including nightlife or entertainment areas, shopping districts, major roads, or schools. In these places, the number of people present can be significantly larger than the number of people who actually live there. It also implies that residential population alone may be an imperfect denominator for measuring crime burden in highly active districts.

V: Killer Plot

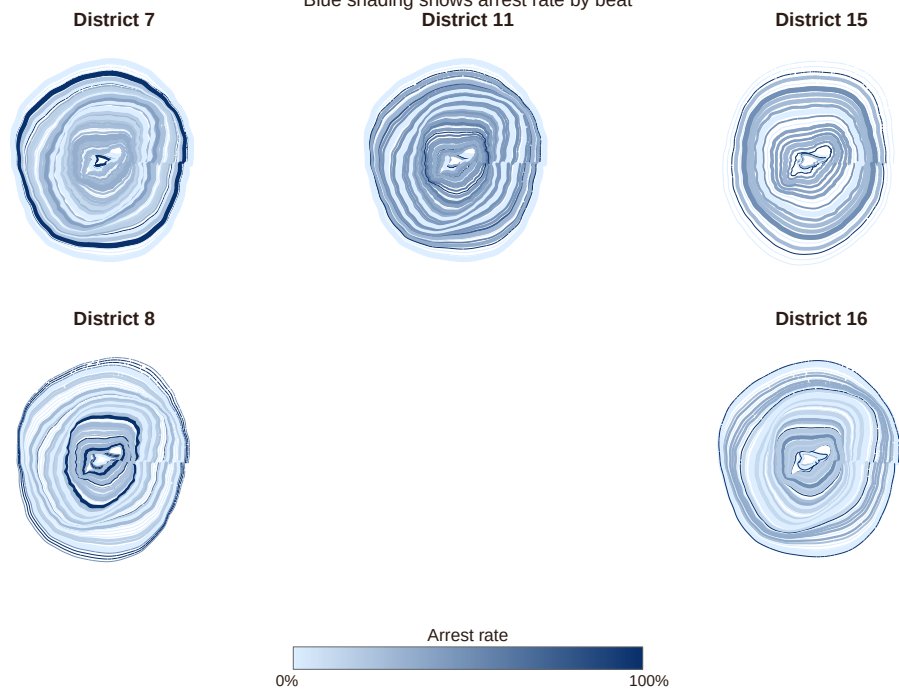
Crime Fingerprints by Police District

Ridges are ordered by beat distance from each district center



Crime Fingerprints by Police District

Blue shading shows arrest rate by beat



This is our killer plot, which we designed around a crime-related image: a fingerprint. Each fingerprint represents a police district, and each ring represents a beat within that district. The waviness of each ring reflects the variation in crime counts across years, so beats with more instability appear less smooth. In the Shiny version, we included a year slider to show how the fingerprints change over time, along with a toggle that switches the shading between violence rate and arrest rate. In this static version shown here, the red plot represents violence rate and the blue plot represents arrest rate.

This plot reinforces what we saw earlier in the district bar charts as we chose to illustrate the districts with the highest and lowest crime rates. Districts 7, 11, and 15 have the highest crime rates, and districts 8 and 16 have the lowest crime rates. This visualization helps us compare not just how much crime occurs in a district, but how crime is distributed across its beats. It is evident that the crime burden is not shared equally across beats in the same district. Furthermore, the violence rate plot suggests that crime count does not necessarily correlate with violence rate, as there is no clear pattern in red intensity in the five districts shown. District 7 seems to be the most violent district, while district 15 appears the least violent. In the arrest rate plot, districts 7 and 8 appear to have significant differences between beats, suggesting that certain beats may be more effective at detecting or responding to crime. In contrast, districts 11, 15, and 16 have more uniform blue shading, which suggests that arrest patterns are more consistent across beats in those districts. Overall, the fingerprint plots emphasize that understanding crime at the district level alone is insufficient, since substantial variation exists within districts and across beats.

Conclusion

This report has analyzed crime trends in Chicago from 2001 to 2025 over a temporal perspective, a “microscopic” (crime-by-crime) perspective, and a community-level geographic analysis. By drawing together statistical analysis tools and secondary geographic and weather data sets, our understanding of Chicago crimes from these three perspectives has deepened. From our analysis, we know that Chicago crimes have been decreasing consistently over time, with consistent and predictable fluctuations/oscillations occurring based on month and seasonal temperatures. The major crimes in Chicago are assault, battery, burglary, criminal damage, theft, motor vehicle theft, and narcotics. These crimes most commonly occur on streets, sidewalks, residences, and apartments; apart from narcotics violations, they generally do not result in an arrest. If these crimes also result in a domestic violence categorization, the odds of an arrest generally increase except in the case of narcotics violation, in which case it decreases. There is a significant disparity between the population and per capita crime rates in certain police districts, which is indicative of deeper systemic issues with the operation of the Chicago police department and how they assign officers to districts and beats. Taken on their own, these insights are interesting for a statistical analysis of trends. However, we have uncovered trends that could be looked into further by sociologists more well-versed in the language of crime and communities in order to understand the systemic reasons for some of these trends and how best to rectify them.

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